

Joint optimization of bidding and pricing strategy for electric vehicle aggregator considering multi-agent interactions

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HIGHLIGHTS

- A novel joint bidding and pricing strategy for EVA is proposed considering the interactions with DSO and EV users.
- The uncertainty of baseline load in the market clearing process is dealt with stochastic optimization.
- A robust semi-dynamic traffic assignment model is built to derive EV charging loads.
- An iterative solution method based on fixed-point theory is designed.

ARTICLE INFO

Keywords:

Electric vehicle aggregator
Bidding and pricing strategy
Semi-dynamic traffic flow
Uncertainty
Bi-level optimization
Robust optimization

ABSTRACT

With the proliferation of electric vehicles (EVs), the EV aggregator (EVA) operating multiple EV charging stations, has emerged to provide charging services to EV users by setting differentiated charging prices to respond to time-varying electricity prices. Existing studies on the profitable strategy of EVA mostly focus on the formulation of bidding or pricing schemes or the optimization of single time-period coordination, failing to fully combine these sequential strategies and exploit the spatial-temporal shifting characteristics of EVs. To address this bottleneck, a novel multi-period joint bidding and pricing strategy for EVA considering the interactions with distribution system operator (DSO) and EV users is formulated. Firstly, a stochastic bi-level bidding and pricing joint optimization model is established, where the market clearing process considering baseline load uncertainties is simulated at the lower level. Secondly, to estimate the dynamic charging behaviors of EV users under differentiated charging prices, a robust semi-dynamic traffic assignment (SDTA) model is constructed to derive EV charging loads considering their coupling effect under traffic restrictions. Finally, an iterative method based on fixed-point theory is designed to obtain the optimal bidding and pricing strategy of EVA, which derives the equilibrium solution by sequentially solving the stochastic bi-level bidding and pricing subproblem and the robust SDTA subproblem. Numerical results verify the effectiveness of the proposed method in obtaining a profitable and stable strategy of EVA when embedded with the practical decision-making process of DSO and EV users.

1. Introduction

With the increasingly serious energy crisis and environmental problems, it has become a common consensus of governments to actively develop electric vehicles (EVs) to replace traditional internal combustion engine vehicles [1,2]. To meet the daily travel demand of EV users, the EV aggregator (EVA), serving as the operator of multiple EV charging stations (EVCSs), is responsible to provide charging services. The profit-driven EVA purchases electricity from day-ahead electricity

market and sets charging price to make arbitrage. Thus, the profit of EVA is influenced by the bidding and pricing strategy. Typically, the EVA needs to predict all charging demands and make an optimal bidding strategy in the day-ahead electricity market. Then, the EVA sets an attractive charging price and schedules EV charging in real time to maximize profit. It should be noted that the deviation between day-ahead energy bids and real-time energy schedule is likely to impose additional penalty cost on the EVA [3]. Hence, it is crucial for EVA to conduct a joint optimization of bidding and pricing strategy to obtain

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<https://doi.org/10.1016/j.apenergy.2024.122810>

Received 17 August 2023; Received in revised form 13 January 2024; Accepted 2 February 2024

Available online 17 February 2024

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the maximum profit.

In the bidding process, the interaction between EVA and distribution system operator (DSO) mainly involves the EVA submitting energy bids to DSO. After collecting all energy bids and supply offers, the DSO conducts market clearing and provides clearing price to EVA. Generally, the EVA participates in the day-ahead electricity market bidding with the aim of minimizing the electricity purchase cost [4,5]. The charging load in each time period is a key factor affecting the power purchase cost of EVA. For the sake of convenience, the EVA usually develops typical scenarios based on historical observations to estimate the schedulable time periods of different EVs and calculates the charging load in each time period according to coordinated charging [6,7]. However, this charging load modeling method is highly dependent on statistical data, especially it cannot reflect the spatial shifting characteristic of EV charging loads [8]. Moreover, the driving pattern of individual EV user has many uncertain factors, such as travel time [9] and the state of charge of batteries [10], which will cause deviations in the EVA's estimation of charging loads. With the development of Vehicle-to-Grid (V2G) technology, EVs can be viewed as mobile energy storage and are no longer just charging loads in power distribution network (PDN). The EVA can use EVs as energy backup devices to obtain additional revenue by supplying energy to the grid [11–13]. The tariff at each timeslot is another key factor that affects the procurement cost of EVA in bidding. In addition to the uncertainty of charging loads, the EVA is confronted with the uncertainty of tariffs [14]. These uncertainties will ultimately make it difficult for EVA to determine an economically feasible bidding strategy in the day-ahead electricity market. A scenario-based stochastic programming method is widely developed to deal with the uncertainties of EV charging loads and tariffs [15]. Since the EVCSs are generally modeled as price-taker, the tariff is assumed as an exogenous parameter [16,17]. However, the suboptimality of bidding strategy that ignores the impact of EVA charging loads on tariffs is shown in [18], even when the penetration of EVs is relatively low, not to mention that the current penetration of EVs has reached a level that cannot be ignored. To briefly reflect the relationship between tariffs and charging loads, a number of endogenous price models are proposed, in which the tariff is modeled as a linear [19] or nonlinear [20] function of charging load. In fact, the relationship between tariffs and charging loads is complex. The market settlement is dependent on the bids of all market participants. Accordingly, in order to make an optimal bidding strategy, the EVA should simulate a real market clearing process. A bi-level bidding strategy of EVA modeled as price-maker is proposed in [21]. However, the market clearing price based on direct current optimal power flow (DCOPF) is not as accurate as that based on alternating current optimal power flow (ACOPF) [22], which reflects the real branch parameters and network loss is considered. In addition, the locational marginal price based on DCOPF can only reflect the regulation demand of distribution system, while the distribution locational marginal price (DLMP) based on ACOPF can reflect the regulation demand of specific nodal load in PDN. In order to achieve the vehicle-grid integration effectively, the EVA needs to fully utilize spatial-temporal shifting characteristic of EVs based on the different tariffs of multiple EVCSs. As a result, the EVA can obtain the optimal bidding strategy during market participation and have the potential to improve the operation of PDN.

In the interaction between EVA and EV users, the EVA continuously optimizes pricing strategy with the goal of maximizing its own benefit. Similar to using a linear function to represent the relationship between market clearing price and charging load during the bidding process, some published researches use demand elasticity to simply approximate the simulation of charging loads under different charging prices [23]. Actually, EV users will determine the charging strategy by minimizing the total charging cost. The shifting cost will be added to the objective function if the spatial flexibility of EV charging is considered [24]. However, these processing methods are too rough to accurately reflect the impact of charging price on the temporal and spatial distribution of

charging load. The charging decisions of EVs are not only related to themselves, but also to other gasoline vehicles (GVs) in transportation network (TN). Accordingly, the traffic assignment (TA) model which can truly reflect the traffic flow is more suitable to precisely characterize the relationship between charging loads and charging prices [25]. In TN, rational users drive from origins to their destinations with the goal of minimizing the travel cost, consist of the cost of travel time and charging service. For fairness, when EV or GV users with the same origin and destination have the same and minimum travel cost, the TN reaches the user equilibrium (UE) state [26]. According to the different time intervals, TA models are subdivided into 3 categories: static TA (STA) model with time interval >90 min [27], semi-dynamic TA (SDTA) model with time interval between 15 and 90 min [28], and dynamic TA (DTA) model with time interval <15 min [29]. As the time interval becomes shorter, TA models become more and more accurate in characterizing the traffic flow dynamics. The STA model is usually used to reflect the steady-state distribution of traffic flow at a single period, while the SDTA and DTA model can provide traffic flow dynamics in a more continuous way over multiple time period. Since the time scale of EVA pricing is usually 1 h, the SDTA model can be utilized to simulate the interaction between EVA and EV users under traffic restrictions. In TA models, the traffic demand from specific origin to destination is usually subject to uncertainty. The uncertainty of traffic demand will inevitably affect the revenue of EVA. In order to ensure the charging demand can be met in the worst traffic demand scenario, robust optimization has emerged as a practicable approach to deal with such uncertainty [30]. Compared with stochastic optimization, the exact probability distribution of traffic demand is not necessary for robust optimization [31], avoiding higher computation cost caused by a large number of scenarios. A robust operation model of PDN coupled with TN is built in [32], taking the upper boundary of the charging load fluctuation range as the worst scenario during the operation of PDN from the perspective of DSO.

To the best of our knowledge, seldom do existing works concentrate on the joint optimization of bidding and pricing strategy for EVA. Most researches focus only on one aspect of bidding and pricing strategy optimization for EVA. Among the few works on the joint optimization of bidding and pricing strategy, a bi-level bidding and pricing strategy from the view of charging network operator is proposed in [33], using a STA model to simulate the relationship between charging load and charging price. In [34], the competitive bidding and pricing problem of single profit-driven EVCS is formulated as tri-level optimization problem with the assumption that there are multiple EVCSs in the region of interest. However, the above researches only focus on the joint optimization of bidding and pricing strategy at a single time period without uncertainties into consideration. Even if the above methods are extended to multiple time periods, the temporal shifting characteristic of charging loads under time-varying charging prices cannot be reflected due to the lack of time coupling constraints. In [35], the EVA calculates a bi-level bidding and pricing strategy considering the uncertainties of EVs and market price by participating in not only electricity market but also carbon market. Nevertheless, the EVA is modeled as a price-taker and the impact of charging loads on market price is neglected.

As stated above, the objective of this paper is mainly to deal with the bidding and pricing problem of EVA in interaction with DSO and EV users. In order to obtain the optimal joint bidding and pricing strategy for EVA, the uncertainties of EV charging loads and tariffs are taken into consideration. The key contributions of this paper are summarized as follows:

- 1) A novel multi-period joint bidding and pricing strategy is proposed to achieve the vehicle-grid integration from the view of a profit-driven EVA, which is modeled as a price-maker when bidding. Especially, the spatial-temporal shifting characteristics of EV charging loads under time-varying charging prices is represented by a SDTA model.
- 2) Multiple uncertainties in the bidding and pricing process are taken into consideration. The uncertainty of traffic demand in TN is dealt

with robust optimization and a robust SDTA model is built for EVA to derive the charging load in each time period while bidding and pricing. In addition, when simulating market clearing to obtain the electricity purchase price, the uncertainty of baseline loads in PDN is handled by stochastic programming.

- 3) In order to obtain the optimal joint bidding and pricing strategy for EVA, an iterative method based on fixed-point theory is proposed to solve the bi-level bidding and pricing subproblem and robust SDTA subproblem in turn.

The remaining parts of this paper are organized as follows. The joint bidding and pricing model structure and the uncertainty modeling of baseline loads in PDN and traffic demand in TN are discussed in Section II. Then, the bi-level bidding and pricing model and robust SDTA model are formulated in Section III. Section IV provides several relaxation and approximation methods and designs an iterative method to obtain the optimal joint bidding and pricing strategy based on fixed-point theory. Case studies are conducted in Section V. Conclusions are given in Section VI.

Nomenclature

Sets

T	Set of time slots
N	Set of EVCSs
W	Set of baseline loads scenarios
Ψ_N, Ψ_L	Set of buses/lines in the PDN
$\pi(j)$	Set of child buses of bus j
$T_N^{R/D}$	Set of origin/destination nodes
$T_A^{R/C/B}$	Set of regular/charging/bypass links in TN, with $T_A = T_A^R \cup T_A^C \cup T_A^B$
$R(j)$	Set of charging links connected to bus j
$K_{r,s}^{g/e}$	Set of available paths for GV/EV between O—D pair (r, s)

Indices

t	Index of time slots
n	Index of EVCS
i, j	Index of buses in PDN, with $j = 0$ denoting root bus
ω	Index of scenarios of PDN baseline loads
a	Index of links in TN
k	Index of paths between O—D pair (r, s) in TN
(r, s)	Index of O—D pairs

Parameters

$\lambda_{n,t}^C, \bar{\lambda}_{n,t}^C$	Charging price boundaries of EVCS n
C_t	Average charging price level of all EVCSs
a_j	Energy production cost coefficients at bus j
r_{ij}, x_{ij}	Resistance and reactance of line between bus i and j
$P_{j,t}^L, Q_{j,t}^L$	Traditional active and reactive power demand at bus j
$v_i, \bar{v}_i$	Bounds of square voltage magnitude at bus i
$P_i^g, \bar{P}_i^g$	Active power generation bounds at bus i
$Q_i^g, \bar{Q}_i^g$	Reactive power generation bounds at bus i
$\bar{l}_{ij}$	Square of current flow capacity of line from bus i to j
c_a^R	Traffic flow capacity of regular link $a \in T_A^R$
c_a^C	Charging service capacity of charging link $a \in T_A^C$
t_a^{R0}	Free travel time on regular link $a \in T_A^R$
t_a^{C0}	Charging time of each EV
P^{ch}	Rated power of the charger
E^B	Energy demand of each EV
η	Charging efficiency

Δt	Duration of each time period
$q_{r,s,t}^{g/e}$	Traffic demand of GV/EV between O—D pair (r, s)
$\delta_{r,s,k,a}^{g/e}$	Link-path incidence matrix element of GV/EV for link a and path k between O—D pair (r, s)
$\zeta_{r,s,k,n}^c$	Correspondence between EVCS n and EV path k between O—D pair (r, s)
w	Monetary cost (\$) of travel time
λ_t^{LMP}	Electricity purchase price of PDN from upper grid

Variables

$\lambda_{n,t}^{\text{DLMP}}$	PDN electricity purchase price (also the corresponding DLMP) at EVCS n
$\lambda_{n,t}^C$	Charging price at EVCS n
$P_{j,t}^g, Q_{j,t}^g$	Active and reactive power generated at bus j
$P_{ij,t}, Q_{ij,t}$	Active and reactive power flow in each line from bus i to j
$P_{j,t}^c$	Power of EV charging loads at bus j
$l_{ij,t}$	Squared magnitude of current in each line from bus i to j
$v_{i,t}$	Squared magnitude of voltage at bus i
$x_{a,t}$	Aggregated travel flow on link a
$t_{a,t}$	Travel time on link a
$f_{r,s,k,t}^{g/e}$	GV/EV flow on path k between O—D pair (r, s)
$t_{r,s,k,t}^{g/e}$	GV/EV travel time on path k between O—D pair (r, s)
$r_{r,s,t}^{g/e}$	GV/EV residual flow between O—D pair (r, s)
$q_{r,s,t}^{g/e, \text{mod}}$	Modified traffic demand of GV/EV between O—D pair (r, s)
$c_{r,s,k,t}^{g/e}$	Travel cost of GV/EV on path k between O—D pair (r, s)
$u_{r,s,t}^{g/e}$	Minimal travel cost of GV/EV between O—D pair (r, s)

2. Model structure and uncertainty modeling

2.1. Framework of EVA joint bidding and pricing problem

This article aims at determining the optimal multi-period joint bidding and pricing strategy to obtain maximum profit from the perspective of EVA, with the SDTA model to represent the relationship between EV charging loads and time-varying charging prices. The illustrative structure of EVA joint bidding and pricing problem is shown in Fig. 1.

In Fig. 1, there are two kinds of interactions related to EVA:

- 1) The responsibility of EVA is to submit the charging demand bids $P_{n,t}^{\text{EV}}$ ($P_{n,t}^{\text{EV}} \geq 0$ and no V2G is considered) of EVCS n when participating in day-ahead electricity market bidding. After collecting all demand bids and supply offers, DSO will conduct market clearing with the goal of minimizing operation cost or maximizing social welfare to obtain the DLMP of each PDN node. It should be pointed out that the electricity purchase price $\lambda_{n,t}^{\text{DLMP}}$ of EVCS n is the DLMP of PDN node connected with EVCS.
- 2) In order to achieve profitability, the EVA needs to set an optimal charging price $\lambda_{n,t}^C$ with given electricity purchase price to provide charging services to EV users in the transportation system. According to the principle of UE, different charging prices will affect the distribution of traffic flow in TN. Thereby, the EVA can obtain maximum profit by continuously adjusting the charging price to change the charging decisions of EV users.

Accordingly, in order to obtain the optimal joint bidding and pricing strategy, the EVA should comprehensively consider these two relationships. In addition, a number of uncertain elements are modeled in this section, which will partly affect the profit of joint bidding and pricing.

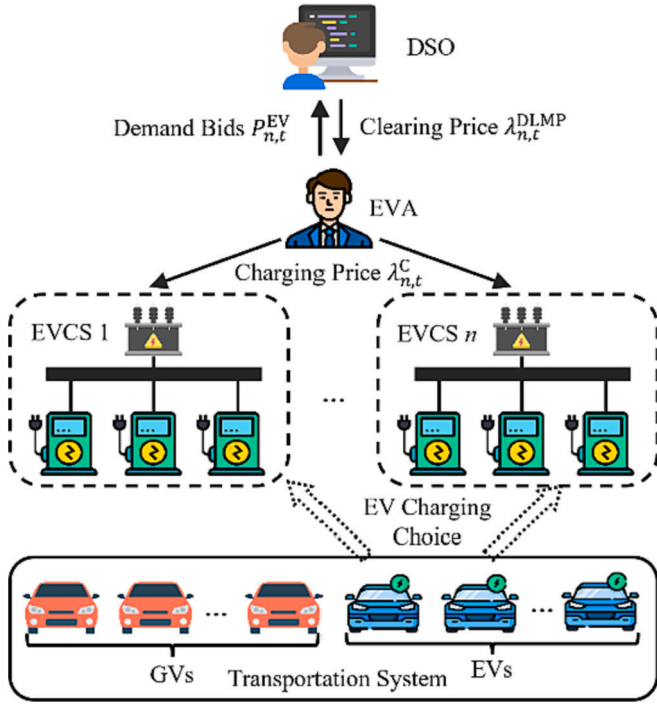


Fig. 1. Structure of EVA joint bidding and pricing problem.

2.2. Uncertainty modeling

In order to determine the optimal joint bidding and pricing strategy, the EVA needs to have complete information on the PDN and TN parameters, i.e., baseline loads in PDN and traffic demand in TN, which may not be exactly known in practice. Therefore, it is necessary to consider the uncertainty of these random parameters while bidding and pricing.

There are two kinds of popular tools to deal with uncertainties. One is the probabilistic optimization method, e.g., stochastic programming (SP), and the other is the optimization method based on the variation interval of uncertainty, such as robust optimization (RO). The main idea of SP is to establish a finite set of scenarios to represent the possible realization of uncertainty. While RO is to find out the optimal decision in the worst scenario within the variation range of uncertainty.

We assume that the EVA deduces the probability distribution of baseline loads in PDN according to the historical information. As a result, we can utilize SP to handle the uncertainty of baseline loads in PDN, which is formulated as:

$$W = \{P^L | P_1^L, P_2^L, \dots, P_\omega^L, \dots, P_{\mathcal{W}}^L\} \quad (1)$$

where the set W includes the possible realization scenarios of the baseline loads in PDN and the total number of all scenarios is $\mathcal{W}$.

The traffic demand at each timeslot is affected by individual driving pattern. Compared with the baseline loads in PDN, the traffic demand in TN has greater randomness. It is difficult for EVA to deduce a specific probability distribution function of traffic demand according to the historical information, while the approximate variation range of traffic demand is easier to be obtained. Hence, we can tackle the uncertainty of traffic demand with RO. However, the results of RO are usually inevitably too conservative [36]. Then, we incorporate budget constraints with an adjustable parameter Γ called uncertainty budget to control the conservatism, which reflect the attitude of operator towards risks. A larger Γ means more conservative decisions. The uncertainty of traffic demand in TN is formulated as:

$$U = \left\{ \begin{array}{l} q_{r,s,t} = \hat{q}_{r,s,t} + z_{r,s,t}^+ \bar{\zeta}_{r,s,t} + z_{r,s,t}^- \underline{\zeta}_{r,s,t} \\ z_{r,s,t}^+ + z_{r,s,t}^- \leq 1, \sum_{r,s,t} (z_{r,s,t}^+ + z_{r,s,t}^-) \leq \Gamma \\ z_{r,s,t}^+ \geq 0, z_{r,s,t}^- \geq 0 \end{array} \right\} \quad (2)$$

where $\hat{q}_{r,s,t}$ is the predicted value of traffic demand between O—D pair (r, s) in period t including GV and EV traffic demand. While $\bar{\zeta}_{r,s,t}$ and $\underline{\zeta}_{r,s,t}$ are the upper and lower boundaries of predicted errors. The variables $z_{r,s,t}^+$ and $z_{r,s,t}^-$ are introduced to control the conservatism of RO.

3. Problem formulation

In order to simulate the real market clearing process, the mathematical formulation of bi-level bidding and pricing model is firstly presented in this section. Then, the robust SDTA model considering the uncertainty of traffic demand is formulated in Section III-B, and the nexus between PDN and TN is modeled in Section III-C.

3.1. Bi-level bidding and pricing model

In this article, the EVA within the region is assumed to be profit-driven. While bidding and pricing, the EVA needs to purchase electricity from day-ahead electricity market at the DLMP obtained by market clearing, and then set an optimal charging price to meet the charging demand of on-road EVs in the transportation system. The clearing price DLMP is determined by the baseline loads and EVCS charging loads. We characterize the uncertainty of PDN baseline loads by constructing a large number of scenarios. Each scenario ω has a probability of occurrence π_ω such that $\sum_{\omega \in W} \pi_\omega = 1$. Hence, the optimal multi-period bidding and pricing problem for EVCSs is formulated as a stochastic bi-level optimization model.

The upper-level model simulates the bidding and pricing of EVA and is as follows:

$$\max \sum_{n \in N} \sum_{t \in T} (\lambda_{n,t}^C - \lambda_{n,t}^{DLMP}) P_{n,t}^{EV} \quad (3a)$$

Subject to:

$$\lambda_{n,t}^C \leq \lambda_{n,t}^C \leq \bar{\lambda}_{n,t}^C \quad (3b)$$

$$\frac{1}{\mathcal{N}} \sum_{n \in N} \lambda_{n,t}^C = C_t \quad (3c)$$

In model (3), the EVA aims to maximize profits by adjusting charging price $\lambda_{n,t}^C$ with given electricity purchase price $\lambda_{n,t}^{DLMP}$ which stems from the solution of the lower-level model. $P_{n,t}^{EV}$ denotes the charging load at EVCS n which is obtained by solving robust SDTA model in TN and is a parameter in this model. Constraints of the upper-level model include box constraints of charging price (3b), and the average charging price level constraint of multiple EVCSs (3c). Let $\mathcal{N}$ denote the number of EVCSs and the average level of charging price C_t is a parameter greater than the average electricity purchase price, ensuring the profitability of the EVA. In order to balance the interests of both EV users and EVA, the average charging price is usually regulated by government departments.

The lower-level model simulates the market clearing process of DSO with the goal of minimizing operation cost considering the uncertainty of baseline loads in PDN based on ACOPF. The electricity purchase price $\lambda_{n,t}^{DLMP}$ of EVCS n is determined by the corresponding DLMP. Generally, the PDN has a radial topology, which can be described as a graph $G_E = [\psi_N, \psi_L]$. The sets of buses and distribution lines are separately represented as ψ_N and ψ_L . The power flow in PDN with a tree topology is usually represented by branch flow equations [37]. After convex relaxation [38], the convexified branch-flow-based ACOPF model can be easily solved by commercial solvers, such as CPLEX. By solving the

lower-level model, the output power of generators, the amount of electricity purchased from upper grid and the DLMPs in each scenario can be obtained. The lower-level model is as follows:

$$\min \sum_{\omega \in W} \sum_{t \in T} \pi_{\omega} \left(\sum_{j \in \Psi_N} a_j P_{j,t,\omega}^E + \lambda_{j,t,\omega}^{\text{LMP}} \sum_{j \in \pi(0)} P_{0j,t,\omega} \right) \quad (4a)$$

Subject to:

$$P_{ij,t,\omega} - r_{ij} l_{ij,t,\omega} + P_{j,t,\omega}^E = \sum_{k \in \pi(j)} P_{jk,t,\omega} + P_{j,t,\omega}^L + P_{j,t}^C : \lambda_{j,t,\omega}^P, \forall j \in \Psi_N, t \in T \quad (4b)$$

$$Q_{ij,t,\omega} - x_{ij} l_{ij,t,\omega} + Q_{j,t,\omega}^E = \sum_{k \in \pi(j)} Q_{jk,t,\omega} + Q_{j,t,\omega}^L, \forall j \in \Psi_N, t \in T \quad (4c)$$

$$v_{i,t,\omega} - v_{j,t,\omega} = 2(r_{ij} P_{ij,t,\omega} + x_{ij} Q_{ij,t,\omega}) - (r_{ij}^2 + x_{ij}^2) l_{ij,t,\omega}, \forall ij \in \Psi_L, t \in T \quad (4d)$$

$$l_{ij,t,\omega} v_{i,t,\omega} \geq P_{ij,t,\omega}^2 + Q_{ij,t,\omega}^2, \forall ij \in \Psi_L \quad (4e)$$

$$P_{ij,t,\omega} - r_{ij} l_{ij,t,\omega} \geq 0, \forall ij \in \Psi_L \quad (4f)$$

$$Q_{ij,t,\omega} - x_{ij} l_{ij,t,\omega} \geq 0, \forall ij \in \Psi_L \quad (4g)$$

$$\underline{v}_i \leq v_{i,t,\omega} \leq \bar{v}_i, \forall i \in \Psi_N \quad (4h)$$

$$0 \leq l_{ij,t,\omega} \leq \bar{l}_{ij}, \forall ij \in \Psi_L \quad (4i)$$

$$\underline{P}_i^E \leq P_{i,t,\omega}^E \leq \bar{P}_i^E, \forall i \in \Psi_N \quad (4j)$$

$$\underline{Q}_i^E \leq Q_{i,t,\omega}^E \leq \bar{Q}_i^E, \forall i \in \Psi_N \quad (4k)$$

In model (4), the objective function (4a) represents the PDN expected operation cost including generation cost of all generators and purchase cost from upper grid. Constraints (4b) and (4c) are nodal active and reactive power balancing conditions, where $P_{j,t,\omega}^L$ and $P_{j,t}^C$ separately denote the baseline load in scenario ω and EVCS charging load at bus j . Specifically, Eqs. (4b) and (4c) guarantee that the injection power plus the power of the generator is equal to the sum of outflow power and total power demand at each bus. It should be noted that $\lambda_{j,t,\omega}^P$ following a colon is the dual variable of constraint (4b) which is the corresponding DLMP at bus j . The electricity purchase price sent to the upper-level model is the sum of DLMP of each probability scenario, which can be expressed as

$$\lambda_{n,t}^{\text{DLMP}} = \sum_{\omega \in W} \pi_{\omega} \lambda_{j,t,\omega}^P \quad (4l)$$

where EVCS n is connected to bus j in PDN.

The voltage drop relationship between adjacent nodes is determined by Eq. (4d). Inequality constraint (4e) defines the apparent power of branch head node after convex relaxation, which has the form of second-order cone. Thus, the lower-level model is actually a tractable second-order cone programming (SOCP). Constraints (4f) and (4g) prevent the occurrence of reverse power flow. Bound constraints of partial decision variables such as squared magnitude of nodal voltage, squared magnitude of line current, active and reactive power output of each generator are imposed in (4h)-(4k).

3.2. Robust SDTA model in TN

As is known that, the bidding and pricing time interval Δt is 1 h. Therefore, the SDTA model is more suitable to simulate the charging decisions of EV users under time-varying charging prices and traffic restrictions for the time interval of SDTA model is between 15 and 90 min. There are already some researches on the model of TN without

considering the uncertainty of traffic demand [39]. The TN is generally represented as a connected graph $G_T = [T_N, T_A]$, where T_N denotes the set of traffic node, including origins, destinations, and intersections; T_A denotes the set of traffic links, such as lanes or roadway segments. Further, links can be divided into three types, i.e., regular links T_A^R , charging links T_A^C and bypass links T_A^B [40]. While regular links represent the physical roads, and the traffic time of vehicles is determined by the degree of road congestion. Charging links refer to EVs passing through EVCSs, and the travel time of which consists of charging time and queuing time, respectively. Bypass links indicate that vehicles pass by the EVCSs and the travel time can be neglected. In TN, vehicles including EVs and GV travel from an origin $r \in T_N^R$ to a destination $s \in T_N^S$ by path $k \in K_{r,s}^{g/c}$. The total number of vehicles traveling from r to s in period t constitutes the initial traffic demand $q_{r,s,t}^{g/c}$. Then the robust SDTA model can be formulated as follows:

$$\max_{q_{r,s,t} \in U} \min_Y \sum_{t \in T} \sum_{r,s} (u_{r,s,t}^g q_{r,s,t}^{g,mod} + u_{r,s,t}^c q_{r,s,t}^{c,mod}) \quad (5a)$$

Subject to:

$$Y = \left\{ f_{r,s,k,t}^g, f_{r,s,k,t}^c, x_{a,t}, t_{a,t}, t_{r,s,k,t}^g, r_{r,s,t}^g, q_{r,s,t}^{g,mod}, t_{r,s,k,t}^c, r_{r,s,t}^c, q_{r,s,t}^{c,mod}, c_{r,s,k,t}^g, c_{r,s,k,t}^c, u_{r,s,t}^g, u_{r,s,t}^c \right\} \quad (5b)$$

$$x_{a,t} = \sum_{r,s} \sum_{k \in K_{r,s}^g} f_{r,s,k,t}^g \delta_{r,s,k,a}^g + \sum_{r,s} \sum_{k \in K_{r,s}^c} f_{r,s,k,t}^c \delta_{r,s,k,a}^c \quad (5c)$$

$$t_{a,t}(x_{a,t}) = t_a^{R0} \left[1 + 0.15 \left(\frac{x_{a,t}}{c_a^R} \right)^4 \right], a \in T_A^R \quad (5d)$$

$$t_{a,t}(x_{a,t}) = t_a^{C0} \left[1 + J \left(\frac{x_{a,t}}{c_a^C - x_{a,t}} \right) \right], a \in T_A^C \quad (5e)$$

$$t_{a,t}(x_{a,t}) = 0, a \in T_A^B \quad (5f)$$

$$t_{r,s,k,t}^g = \sum_a t_{a,t}(x_{a,t}) \delta_{r,s,k,a}^g \quad (5g)$$

$$r_{r,s,t}^g = \sum_{k \in K_{r,s}^g} \frac{f_{r,s,k,t}^g t_{r,s,k,t}^g}{\Delta t} \quad (5h)$$

$$q_{r,s,t}^{g,mod} = q_{r,s,t}^g + \frac{1}{2} r_{r,s,t}^g - \frac{1}{2} r_{r,s,t}^c \quad (5i)$$

$$\sum_{k \in K_{r,s}^g} f_{r,s,k,t}^g = q_{r,s,t}^{g,mod} \quad (5j)$$

$$t_{r,s,k,t}^c = \sum_a t_{a,t}(x_{a,t}) \delta_{r,s,k,a}^c \quad (5k)$$

$$r_{r,s,t}^c = \sum_{k \in K_{r,s}^c} \frac{f_{r,s,k,t}^c t_{r,s,k,t}^c}{\Delta t} \quad (5l)$$

$$q_{r,s,t}^{c,mod} = q_{r,s,t}^c + \frac{1}{2} r_{r,s,t}^c - \frac{1}{2} r_{r,s,t}^g \quad (5m)$$

$$\sum_{k \in K_{r,s}^c} f_{r,s,k,t}^c = q_{r,s,t}^{c,mod} \quad (5n)$$

$$c_{r,s,k,t}^g = w t_{r,s,k,t}^g, \forall k \in K_{r,s}^g \quad (5o)$$

$$c_{r,s,k,t}^c = w t_{r,s,k,t}^c + \lambda_{n,t}^C \zeta_{r,s,k,n} E^B, \forall k \in K_{r,s}^c \quad (5p)$$

$$0 \leq f_{r,s,k,t}^g \perp c_{r,s,k,t}^g - u_{r,s,t}^g \geq 0 \quad (5q)$$

$$0 \leq f_{r,s,k,t}^c \perp c_{r,s,k,t}^c - u_{r,s,t}^c \geq 0 \quad (5r)$$

$$(2) \quad (5s)$$

In model (5), the objective function (5a) represents the minimum travel cost of all EVs and GVs under the worst case considering the varied traffic demand. Eq. (5c) defines the relation between link flow $x_{a,t}$ and path flow, where $f_{r,s,k,t}^g$ and $f_{r,s,k,t}^c$ represent the GV and EV flow on path k between each O—D pair, respectively. The travel time of regular link, charging links and bypass links are expressed as (5d)–(5f). While (5d) is the most widely used travel time function of regular link flow named Bureau of Public Roads (BPR) function [41]. Physically, the regular link capacity is the traffic flow when $t_{a,t} = 1.15t_a^0$ and is not a mandatory limit. When $x_{a,t} > c_a^R$, the travel time of regular link will rapidly increase. Constraint (5e) is named Davison function based on the queuing theory, consist of charging time $t_a^{co} = E^B / (P^{ch}\eta)$ and queuing time $J t_a^{co} x_{a,t} / (c_a^c - x_{a,t})$ [42]. Different from previous researches [33,40], the charging efficiency is taken into account in this article while calculating the charging time. Let η represent the charging efficiency. Parameters E^B and P^{ch} denote the charging energy and rated power, respectively. When $x_{a,t}$ tends to c_a^c , the queuing time will approach to infinity. The parameter J controls the shape of the function. Constraint (5f) indicates the travel time of bypass links can be neglected due to the short length of bypass links.

Eq. (5g) represents the travel time of GVs on path k between each O—D pair. Eqs. (5h)–(5j) and (5l)–(5n) reflect the typical characteristics of the SDTA model, compared with the STA model. We suppose that the travel time of each path is no greater than Δt and the departure rate of drivers is uniform during each time period [28]. Hence, there will be part of traffic flows that fail to reach the destination within the current time period, which is called residual flow and is defined in (5h). The existence of residual flow fully reflects the temporal shifting characteristic of traffic demand. Constraint (5i) represents the modified traffic demand in period t , which includes the initial traffic demand and part of the residual flow of previous time period, and where part of the residual flow of current traffic demand is subtracted and added to the following traffic demand. The modified ratio of residual flow is set to 0.5 in this article for simplicity. Eq. (5j) indicates that the total traffic flow of all paths is equal to the modified traffic demand between each O—D pair in period t . Constraints (5h)–(5j) indicate the modification of traffic demand of GVs based on the SDTA theory. The travel time of EVs on path k between each O—D pair is represented in (5k). Similar to the modification of traffic demand of GVs, the traffic demand of EVs also needs to be modified and is shown in (5l)–(5n).

Eqs. (5o) and (5p) pertain to the travel cost of GVs and EVs on path k , respectively. If each driver chooses the driving path rationally under the guidance of accurate navigation, the traffic flow distribution in TN will reach a Wardrop UE state [26]. It should be noted that the Wardrop UE state is the most beneficial state for GV and EV users. According to the Wardrop UE principle, all paths selected by GV or EV users between each O—D pair should have the same and minimal travel cost. It means no one can reduce the travel cost by changing the driving path. As a result, such constraints can be described as complementary and slackness conditions as shown in (5q)–(5r).

It is worth noting that the robust SDTA model is nonlinear and non-convex. There are bi-linear terms in (5a), (5h) and (5l), quartic monomial term in (5d) and nonlinear function in (5e). In addition, the complementary and slackness conditions shown in (5q) and (5r) also constitute the nonlinearity of the robust SDTA model. All these elements make it impossible for the robust SDTA model to be directly solved by off-the-shelf solvers like Gurobi and CPLEX.

3.3. Modeling of the nexus between PDN and TN

We assume that each EVCS is powered by a bus in PDN, and the

charging load is related to the traffic flow of charging link corresponding to EVCS. The charging load is given by

$$P_{j,t}^c = \sum_{a \in R(j)} x_{a,t} E^B / \eta \quad \forall j \in \psi_N, t \in T \quad (6)$$

4. Solution methods

We note that the robust SDTA model is nonlinear and non-convex, including bi-linear terms in (5a), (5h) and (5l); quartic monomial term in (5d) and nonlinear function in (5e), which brings challenges to solving this model. In this section, we first present three convex relaxation methods to handle these nonlinear and non-convex terms, as well as the approximation method for the complementary and slackness conditions in (5q)–(5r). Then, we present the solution method of the bi-level bidding and pricing model in PDN and the convexified robust SDTA model in TN, respectively. Finally, we design an iteration method to solve the two models based on fixed-point theory.

4.1. Convex relaxation and approximation methods in TN

1) *McCormick Envelope of Bi-Linear Term*: The bi-linear terms in the robust SDTA model can be convexified by their McCormick envelopes. Define the auxiliary variable φ to represent the convexified approximation of bi-linear term xy with upper and lower boundaries $x_{\min} \leq x \leq x_{\max}$ and $y_{\min} \leq y \leq y_{\max}$. The McCormick envelope of the above bi-linear term is formulated as follows:

$$\begin{cases} \varphi \geq xy_{\min} + x_{\min}y - x_{\min}y_{\min} \\ \varphi \geq xy_{\max} + x_{\max}y - x_{\max}y_{\max} \\ \varphi \leq xy_{\min} + x_{\max}y - x_{\max}y_{\min} \\ \varphi \leq xy_{\max} + x_{\min}y - x_{\min}y_{\max} \end{cases} \quad (7)$$

Since there are four bi-linear terms in the robust SDTA model, let auxiliary variables $\lambda_{r,s,t}^g$ and $\lambda_{r,s,t}^c$ denote the convexified approximation of bi-linear terms $u_{r,s,t}^g q_{r,s,t}^{g,mod}$ and $u_{r,s,t}^c q_{r,s,t}^{c,mod}$, respectively. While the auxiliary variables $\kappa_{r,s,k,t}^g$ and $\kappa_{r,s,k,t}^c$ are introduced to represent the convexified approximations of $f_{r,s,k,t}^g t_{r,s,k,t}^g$ and $f_{r,s,k,t}^c t_{r,s,k,t}^c$, respectively.

2) *Convex Hull of Quartic Monomial Term*: There is a quartic monomial term $x_{a,t}^4$ in BPR function (5d), which can be convexified by twice convex hull relaxation for $x_{a,t}^4 = (x_{a,t}^2)^2$. Suppose a continuous variable x with upper and lower boundaries $x_{\min} \leq x \leq x_{\max}$. Define the auxiliary variable χ and χ' to represent the convex hull of the quadratic monomial term x^2 and the quartic monomial term x^4 . The mathematical formulation is as follows:

$$\begin{cases} \chi \geq x^2 \\ \chi \leq (x_{\min} + x_{\max})x - x_{\min}x_{\max} \end{cases} \quad (8a)$$

$$\begin{cases} \chi' \geq \chi^2 \\ \chi' \leq (x_{\min}^2 + x_{\max}^2)\chi - x_{\min}^2 x_{\max}^2 \end{cases} \quad (8b)$$

3) *Piece-Wise Linear Approximation Method of the Nonlinear Term*: In addition to the bi-linear terms and BPR function, the Davison function (5e) also includes a nonlinear term, which can be linearized by piece-wise linear approximation method and is formulated as follows:

$$\begin{cases} x_{a,t} = \sum_h x_a^H \theta_{a,h,t}, a \in T_A^C \\ t_{a,t}(x_{a,t}) = \sum_h t_a(x_a^H) \theta_{a,h,t}, a \in T_A^C \\ \sum_h \theta_{a,h,t} = 1, a \in T_A^C, \{\theta_{a,h,t} \forall h\} \in \text{SOS2} \\ 0 \leq \theta_{a,h,t} \leq 1, a \in T_A^C \end{cases} \quad (9)$$

where x_a^H denote the segmenting points, the selection of which can be either evenly distributed for simplicity or be optimized using the method proposed in [43]. Special ordered set SOS2 $\{\theta_{a,h,t}, \forall h\}$ means there are at most two adjacent elements that are not zero and the others are zero.

4) *Big-M Approximation Method of Complementary and Slackness Conditions*: The complementary and slackness conditions can be converted as the linear constraints by the big-M method. We introduce a binary variable $v_{r,s,t}$ and a large enough constant M to reformulate the complementary and slackness conditions as follows (for symbol brevity, here we omit the subscript g and e for GVs and EVs):

$$\begin{cases} 0 \leq f_{r,s,k,t} \leq M(1 - v_{r,s,t}) \\ 0 \leq c_{r,s,k,t} - u_{r,s,k} \leq Mv_{r,s,t} \end{cases} \quad (10)$$

Incorporating the above-discussed convex relaxation and approximation methods, the robust SDTA model can be transformed into a convex mixed-integer quadratically constrained quadratic program (MIQCQP) problem.

In order to evaluate the performance of the convex relaxation and approximation methods, we rely on the concept of mean absolute percentage error to define several metrics.

$$V_{r,s,t}^{g/e} = \frac{|\lambda_{r,s,t}^{g/e} - u_{r,s,t}^{g/e} q_{r,s,t}^{g/e,mod}|}{u_{r,s,t}^{g/e} q_{r,s,t}^{g/e,mod}} \quad (11a)$$

$$V_{r,s,k,t}^{g/e} = \frac{|\kappa_{r,s,k,t}^{g/e} - f_{r,s,k,t}^{g/e} p_{r,s,k,t}^{g/e}|}{f_{r,s,k,t}^{g/e} p_{r,s,k,t}^{g/e}} \quad (11b)$$

$$V_{a,t} = \frac{|\lambda'_{a,t} - x_{a,t}^A|}{x_{a,t}^A}, a \in T_A^R \quad (11c)$$

$$V_{a,t}^C = \frac{|l_{a,t} - l_a^{C0}(1 + Jx_{a,t}/c_a^C - x_{a,t})|}{l_a^{C0}(1 + Jx_{a,t}/c_a^C - x_{a,t})}, a \in T_A^C \quad (11d)$$

Then the following criteria can be calculated as the average percentage difference between the auxiliary variables and their replaced non-convex and nonlinear terms.

$$V_{\lambda}^{g/e} = 100 \times \frac{\sum_{t \in T} \sum_{r,s} V_{r,s,t}^{g/e}}{|T|N^{g/e}} \quad (12a)$$

$$V_{\kappa}^{g/e} = 100 \times \frac{\sum_{t \in T} \sum_{r,s} \sum_{k \in K_{r,s}^{g/e}} V_{r,s,k,t}^{g/e}}{|T|N^{g/e} |K_{r,s}^{g/e}|} \quad (12b)$$

$$V_a = 100 \times \frac{\sum_{t \in T} \sum_{a \in T_A^R} V_{a,t}}{|T| |T_A^R|} \quad (12c)$$

$$V_a^C = 100 \times \frac{\sum_{t \in T} \sum_{a \in T_A^C} V_{a,t}^C}{|T| |T_A^C|} \quad (12d)$$

where parameter $N^{g/e}$ denotes the number of O—D pairs of GVs/EVs.

4.2. Solution method of the bi-level bidding and pricing model based on strong duality for SOCP

A common way to solve the bi-level optimization model is to introduce the Karush-Kuhn-Tucker (KKT) conditions of the lower-level model into the upper-level model. It will not bring computational challenge if the lower-level model is a linear programming. However, the lower-level model of the proposed bi-level bidding and pricing model is an ACOPF model in SOCP, the KKT conditions of which will introduce huge number of bi-linear terms, making the model difficult to solve.

Another practicable approach to handle the lower-level model in SOCP is to eliminate the complementary and slackness conditions owing to the strong duality of convex SOCP. Ref [44] demonstrated that minimizing the duality gap, which is the difference between the primal and dual objectives, is theoretically equivalent to the complementary and slackness conditions. Therefore, we define a small positive constant δ to restrict the duality gap, and the proposed bi-level bidding and pricing model is equivalent to a single-level model as follows:

$$\begin{aligned} & \max \sum_{t \in T} \sum_{n \in N} (\lambda_{n,t}^C - \lambda_{n,t}^{DLMP}) P_{n,t}^{EV} \\ & \text{s.t. (3b) - (3c)} \\ & \text{primal constraints of lower level model (4b) - (4k)} \\ & \text{dual constraints of lower level model} \\ & A - D \leq \delta \end{aligned} \quad (13)$$

where A and D separately denote the objective of primal and dual model of the lower-level model.

4.3. An altering optimization procedure method for RO with mixed integer recourse

The robust SDTA model proposed in this article is a bi-level max-min RO problem. If the second level of the RO problem only has continuous variables, a single-level maximization problem can be obtained based on strong dual theory or KKT conditions. However, the second level of the proposed robust SDTA model has binary variables introduced by big-M method, resulting in the strong dual theory or KKT conditions no longer applicable. A nested column-and-constraint iteration algorithm to handle the problem by enumerating the binary variables is proposed in [45], which may bring computational complexity. The maximum number of iterations will reach 2^n when there are n binary variables in the second level of the RO problem. In order to enhance the solving efficiency, an altering optimization procedure (AOP) method is used to solve the proposed robust SDTA model with mixed integer recourse. The convergency proof is AOP method can be found in [46].

For simplicity, we first provide a matrix representation of the robust SDTA model, which is formulated as follows:

$$\begin{aligned} & \max_{\mathbf{u}} \min_{\mathbf{y}, \mathbf{z}} \mathbf{y}^T \\ & \text{s.t. } \mathbf{P}\mathbf{y} + \mathbf{Q}\mathbf{z} \leq \mathbf{G}\mathbf{u} + \mathbf{f} \end{aligned} \quad (14)$$

where $\mathbf{y}$ and $\mathbf{z}$ denote the continuous variables and binary variables. While $\mathbf{u}$ represents the uncertainty in the model.

The main idea of AOP method is to solve optimization of $\mathbf{u}$ -fixed subproblem and $\mathbf{z}$ -fixed subproblem alternately, and is presented in Algorithm 1.

1) $\mathbf{u}$ -fixed subproblem:

$$\begin{aligned} & \min_{\mathbf{y}, \mathbf{z}} \mathbf{y}^T \\ & \text{s.t. } \mathbf{P}\mathbf{y} + \mathbf{Q}\mathbf{z} \leq \mathbf{G}\mathbf{u}^* + \mathbf{f} \end{aligned} \quad (15)$$

where $\mathbf{u}^*$ represents a possible realization scenario of the uncertainty, which is set as constant in the $\mathbf{u}$ -fixed subproblem.

2) $\mathbf{z}$ -fixed subproblem:

The bi-level max-min RO model can be transformed into a single-level max model based on the strong dual theory.

$$\begin{aligned} & \max_{\lambda} \lambda^T (\mathbf{G}\mathbf{u} + \mathbf{f} - \mathbf{Q}\mathbf{z}^*) \\ & \text{s.t. } \mathbf{P}^T \lambda \leq \mathbf{c} \\ & \lambda \leq 0 \end{aligned} \quad (16)$$

where $\mathbf{z}^*$ represents the constant and λ denote the dual variables of the second level model after $\mathbf{z}$ is fixed to $\mathbf{z}^*$. However, the objective function of $\mathbf{z}$ -fixed subproblem includes a bi-linear term $\lambda^T \mathbf{u}$, which can be linearized using big-M method [47].

Algorithm 1. AOP

Input:	Basic parameters of TN, charging price of EVCSs $\{\lambda_{n,t}^C\}$, traffic demand predictions $\hat{q}_{r,s,t}$, upper/lower boundaries of predicted errors $\bar{\zeta}_{r,s,t}/\underline{\zeta}_{r,s,t}$, uncertainty budget Γ .
Output:	The worst scenario $\mathbf{u}^*$
1:	Initialization: $r \leftarrow 1$. Initialize the value for $\mathbf{u}_r$.
2:	Repeat
3:	Solve $\mathbf{u}$ -fixed subproblem with $\mathbf{u}_r$ to get $\mathbf{y}_r$ and $\mathbf{z}_r$
4:	Solve $\mathbf{z}$ -fixed subproblem with $\mathbf{z}_r$ to get $\mathbf{u}_{r+1}$
5:	$r \leftarrow r+1$
6:	Until $\mathbf{u}_{r+1} = \mathbf{u}_r$
7:	Return $\mathbf{u}^* = \mathbf{u}_{r+1}$

4.4. Fixed-point view of the joint bidding and pricing problem and the corresponding iteration method

The iterative relations between the EVA, PDN and TN is presented in Fig. 2. There are three optimal decision process described as the links between each box in Fig. 2, which is formulated in Section III, and the boxes represents the key decision variables of the three optimization models. The goal is to obtain the optimal joint bidding and pricing strategy, determining the charging loads of EVCSs in each time period, that makes the three optimization models reach optimality simultaneously.

We first define three mappings to represent the three optimization models as follows:

- 1) $\{\lambda_{n,t}^{\text{DLMP}}\} = M_{\text{OPF}}(\{P_{n,t}^{\text{EV}}\})$: Mapping M_{OPF} stands for the lower-level model of bi-level bidding and pricing model in PDN representing the market clearing process of DSO. As long as the PDN is strong enough, M_{OPF} is feasible for any given charging load distribution $\{P_{n,t}^{\text{EV}}\}$. In addition, the piecewise continuity of DLMP is described in [48] with the change of load owing to the possible changes of binding limit. Hence, M_{OPF} can be assumed to be continuous under the condition that the charging load variations are very small and will not cause the change of binding limit in iterations.
- 2) $\{\lambda_{n,t}^C\} = M_{\text{OP}}(\{\lambda_{n,t}^{\text{DLMP}}\})$: With the DLMP of nodes where EVCSs are located given, mapping M_{OP} standing for the upper-level model of bi-level bidding and pricing model in PDN represents the process of EVA setting the optimal charging price to maximize profit. The feasible region of mapping M_{OP} is fixed and related only to the bounds of the charging price and the average charging price level. We assume that mapping M_{OP} is continuous for the charging price and profit will not change significantly owing to a minor variation in electricity purchase price.

- 3) $\{P_{n,t}^{\text{EV}}\} = M_{\text{UE}}(\{\lambda_{n,t}^C\})$: According to the robust SDTA model, mapping M_{UE} determines the traffic flows of TN with respect to the charging price of EVCSs $\{\lambda_{n,t}^C\}$. The feasibility of mapping M_{UE} is guaranteed as long as TN is strongly connected and the total charging capacity is adequate enough to meet all charging demands. Accordingly, when the charging load is not very heavy, we can assume that mapping M_{UE} is continuous for the linear constraints on charging price in the robust SDTA model.

Then, we define the a synthetical mapping M_{whole} to describe the loop in Fig. 2, which is formulated as follows:

$$\{P_{n,t}^{\text{EV}}\} = M_{\text{whole}}(\{P_{n,t}^{\text{EV}}\}) = M_{\text{UE}}(M_{\text{OP}}(M_{\text{OPF}}(\{P_{n,t}^{\text{EV}}\}))) \quad (17)$$

The charging power of each EVCS is limited by the upper and lower boundaries, for each EVCS has a definite upper bound of charging capacity. Hence, the bidding and pricing space of EVA is defined as

$$\mathcal{F} = \{P_{n,t}^{\text{EV}} | 0 \leq P_{n,t}^{\text{EV}} \leq c_a^C E^B\} \quad (18)$$

We can find that M_{whole} is a self-mapping $M_{\text{whole}}: \mathcal{F} \rightarrow \mathcal{F}$. In addition, M_{whole} is also continuous owing to the continuity assumptions of mappings M_{OPF} , M_{OP} and M_{UE} . The existence of optimal solution to a self-mapping with convex and compact feasible region based on Brouwer fixed-point theory is proven in [49].

Iterative methods can effectively solve the fixed-point problem. In the process of EVA bidding and pricing, for each charging price given by EVA, a specific charging load distribution result can be obtained by solving robust SDTA model. Thus, we can decouple PDN and TN and design an iterative solution method to solve the problem of EVA optimal joint bidding and pricing. The detailed algorithm is described in Algorithm 2.

Just like the problems encountered in other researches [33,40], Algorithm 2 cannot be mathematically proven to converge theoretically, because the mappings we defined are based on optimization problem, instead of mathematical properties. To improve the convergence of the Algorithm 2 and avoid possible oscillations in iterations, we introduce parameter Ω to limit the change in the charging price during two

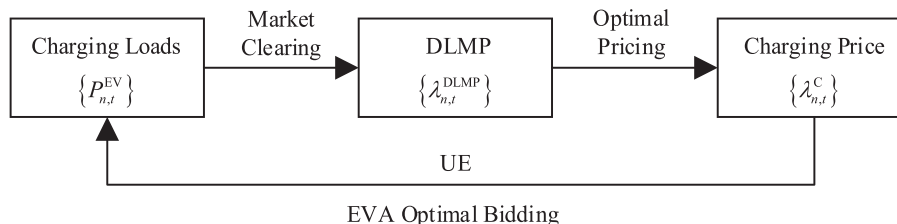


Fig. 2. A fixed-point view of EVA optimal joint bidding and pricing problem.

successive iterations. Suppose $\{\lambda_{n,t}^{\text{DLMP}}(k)\}$ is the electricity purchase price in the k -th iteration and $\{\lambda_{n,t}^{\text{C}}(k)\} = M_{\text{OP}}(\{\lambda_{n,t}^{\text{DLMP}}(k)\})$, the updated charging price can be defined as

$$\lambda_{n,t}^{\text{C}}(k) = \Omega \lambda_{n,t}^{\text{C}}(k) + (1 - \Omega) \lambda_{n,t}^{\text{C}}(k-1) \quad (19)$$

where $\Omega \in [0,1]$ is a hyper-parameter, which can adjust the smoothness of convergence and calculation time. Generally, a smaller Ω can effectively alleviate the oscillations in iterations, but will require more calculation time and iterations to reach the optimal solution. If Algorithm 2 fails to converge within the maximum number of iterations, we can evaluate the results of each iteration and choose the one with the largest joint bidding and pricing profit as the suboptimal solution.

Algorithm 2. Iteration Algorithm for the EVA Optimal Joint Bidding and Pricing

Input: Input of Algorithm 1, Basic parameters of the PDN, scenario set of PDN baseline loads, a convergence tolerance $\varepsilon > 0$, maximal number of iterations K

Output: EVCSs charging load $\{P_{n,t}^{\text{EV*}}\}$ and charging price $\{\lambda_{n,t}^{\text{C*}}\}$

- 1: **Initialize:** $k \leftarrow 1$. Initialize the charging price of EVCSs for $\lambda_{n,t}^{\text{C}}(1) = C_{n,t}$. Set the initial traffic flow $x_{a,t}(1) = 0, a \in T_a^{\text{C}}$.
- 2: **Repeat**
- 3: Solve the robust SDTA model with $\lambda_{n,t}^{\text{C}}(k)$ to obtain the traffic flow distribution and calculate the charging load in PDN $P_{j,t}^{\text{C}}(k+1) = \sum_{a \in W(j)} x_{a,t}(k+1) E^{\text{B}}/\eta$. Then, send the charging load to DSO
- 3: Solve the bi-level bidding and pricing model to obtain the updated charging price $\lambda_{n,t}^{\text{C}}(k+1)$
- 4: $k \leftarrow k+1$. Calculate $\Delta^{\text{TN}} = \|x_{a,t}(k+1) - x_{a,t}(k)\|$ and $\Delta^{\text{PDN}} = \|\lambda_{n,t}^{\text{C}}(k+1) - \lambda_{n,t}^{\text{C}}(k)\|$
- 5: **Until** $\Delta^{\text{TN}} + \Delta^{\text{PDN}} \leq \varepsilon$ or $k > K$
- 6: **If** $k \leq K$ **then**
- 7: Calculate the charging load of EVCSs $P_{n,t}^{\text{EV}}(k+1)$ according to the charging load in PDN $P_{j,t}^{\text{C}}(k+1)$
- 8: **Return** $\{P_{n,t}^{\text{EV*}}\} = \{P_{n,t}^{\text{EV}}(k+1)\}$ and $\{\lambda_{n,t}^{\text{C*}}\} = \{\lambda_{n,t}^{\text{C}}(k+1)\}$
- 9: **Else**
- 10: The EVA evaluates all demand bids $\{P_{n,t}^{\text{EV}}(k)\} (k = 2, 3, \dots, K+1)$ and selects the one with the maximum profit and return the value $\{P_{n,t}^{\text{EV*}}\}$ and corresponding charging pricing $\{\lambda_{n,t}^{\text{C*}}\}$
- 11: **End if**

5. Case studies

5.1. Simulation setup

In this article, we adopt a simple integrated TN-PDN test system (test system 1) and a medium-scale TN-PDN test system (test system 2) to verify the effectiveness and practicability of the proposed model. Due to the low traffic flow in TN at night, EVCSs can only earn low profit by providing charging services. In order to save operational cost, EV charging service time is assumed to span from 8:00 to 18:00. The upper and lower limits for charging price of each EVCS are set to be 200 \$/MWh and 160 \$/MWh, respectively. In addition, the average charging price level of all EVCSs $C_t = 180$ \$/MWh.

1) *Parameters of Test System 1:* The topology of TN in test system 1

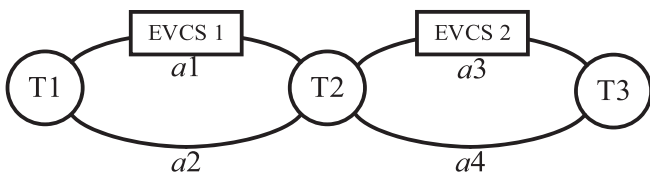


Fig. 3. Topology of 3-node transportation system.

Table 1
Parameters in TN infrastructure.

Link	$t_a^{\text{RO}}(\text{min})$	$c_a^{\text{R}}(\text{p.u.})$
a1	8	45
a2	10	35
a3	11.5	35
a4	8	50

including a single O—D pair (from T1 to T3) with 4 links is shown in Fig. 3. The simple 3-node transportation system is revised from [39]. There are two EVCSs in TN equipped in Link 1 and link 3. For each EV user, the rated charging power and charging demand are set as 150 kW and 50 kWh, respectively. The charging efficiency $\eta = 0.95$. The base value of traffic rate is set as 50 vehicles/h and the charging capacity of each EVCS is $c_a^{\text{C}} = 3$. We assume that the penetration rate of EV is 10% and half of them need recharging. This assumption will result in the actual traffic demand of EV between each O—D pair accounting for only

5% of the overall initial traffic demand. The other parameters of TN are summarized in Table 1. In addition, a 7-bus radial distribution system is coupled with the 4-link transportation system in test system 1. The topology of PDN and connections between PDN and EVCSs are shown in Fig. 4 with line data provided beside each branch [40]. The base value of power is 5 MVA and the current flow capacity of each line is 4. The upper and lower voltage limits of each bus are set to be 1.05 and 0.9, while the voltage of slack bus is 1.04. The two generators in PDN share the same data: $P_i^{\text{S}} = 1, Q_i^{\text{S}} = 0.8, \underline{Q}_i^{\text{S}} = -0.8, a_j = 140$ \$/MWh. The contract price

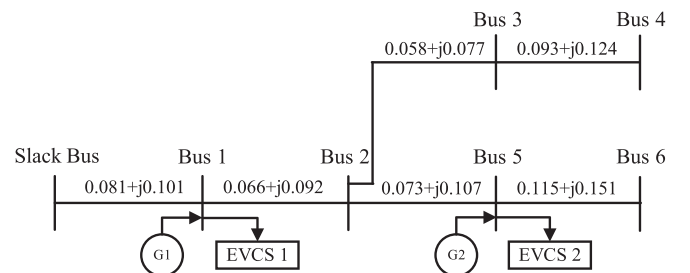


Fig. 4. Topology of 7-bus radial distribution system with line data provided beside each branch.

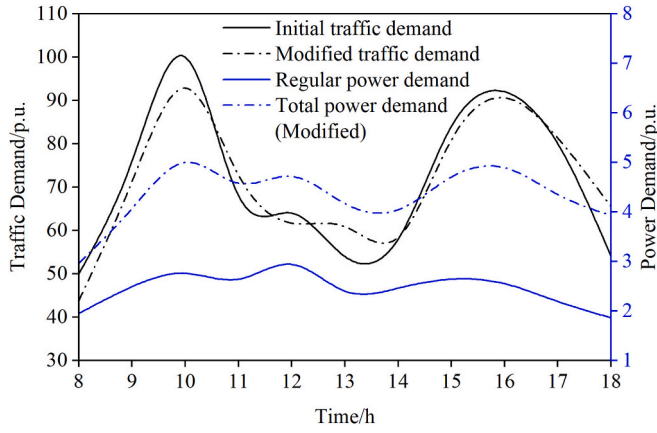


Fig. 5. Daily traffic demand and PDN loads.

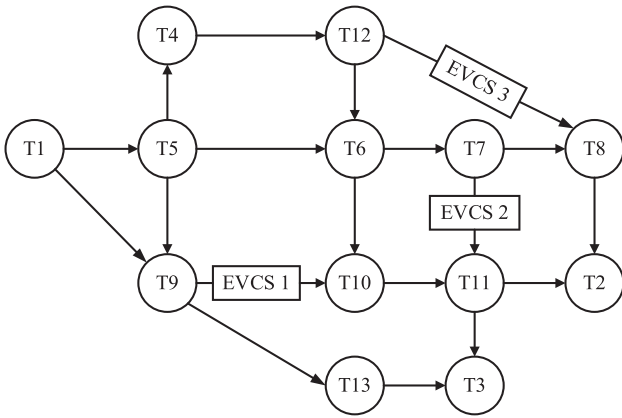


Fig. 6. Topology of 13-node Nguyen-Dupuis transportation system.

with the upper grid is $\lambda_t^{\text{LMP}} = 150$ \$/MWh. The predicted baseline loads in PDN and traffic demand in TN referenced from [50] are shown in Fig. 5. We suppose that the total regular power demand is evenly distributed from bus 1 to bus 6.

2) *Parameters of Test System 2:* In test system 2, a modified Nguyen-Dupuis transportation system [51] consisting of 13 nodes and 19 links is shown in Fig. 6. Here, the TN includes three EVCSs and two O—D pairs (from T1 to T2 and T1 to T3). Similarly, the base values of traffic flow and power demand are 50 vehicles/h and 5 MVA. The link capacities of all regular links are 25, and the charging capacity of each EVCS is set as 2. In addition, the free travel time of each regular link is $t_a^{\text{RO}} = 5$ min. The topology of revised 18-bus radial distribution system is shown in Fig. 7. The impedance and the current flow capacity of each branch have been reported elsewhere [52]. We add three generators at bus 3, 5 and 12. The three generators share the same data: $\bar{P}_i^g = 1$, $\bar{Q}_i^g = 0.8$, $\underline{Q}_i^g = -0.8$. The predicted baseline loads in PDN and traffic demand in TN referenced from [28] are shown in Fig. 8. Other parameters, such as EV parameters, the voltage limits and energy production cost coefficient of power generators, are the same as those of test system 1.

In order to verify the effectiveness of the proposed model and iterative method, we design several cases.

1) *Case I:* Traditionally, the EVA usually utilizes demand elasticity model to represent the relationship between charging loads and time-varying charging prices while bidding and pricing. We assume that the charging demand in each time period will be evenly distributed when all EVCSs share the same charging price, i.e., $\lambda_{it}^{\text{C}} =$

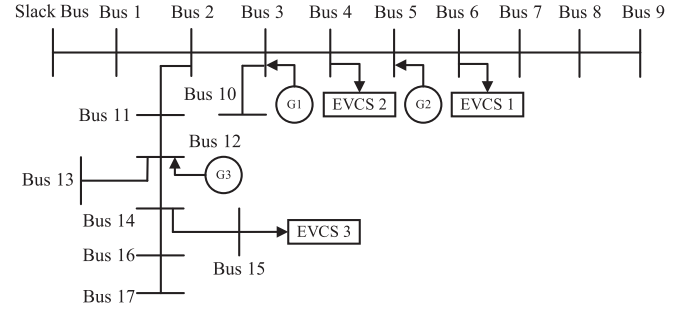


Fig. 7. Topology of 18-bus radial distribution system.

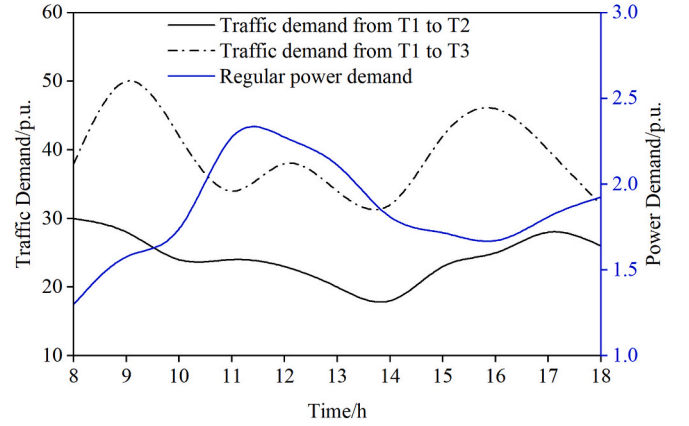


Fig. 8. Profiles of baseline loads in PDN and traffic demand in TN in test system 2.

$C_t = 180$ \$/MWh. For every 10 \$ per MWh increase in charging price, the charging demand will be reduced by 10%. In addition, the uncertainties of baseline loads in PDN and traffic demand in TN are not taken into account.

- 2) *Case II:* According to the proposed bidding and pricing strategy, the EVA accurately simulates the relationship between charging loads and time-varying charging prices during multiple time periods based on the SDTA model in the deterministic case. In addition, the duality gap δ is 0.005, and the tolerance value ϵ is 1e-6.
- 3) *Case III:* The EVA takes the uncertainty of baseline loads in PDN into account while bidding and pricing, which is dealt with SO. We assume that the prediction error of PDN baseline loads is $\pm 10\%$ of the predicted value and the baseline loads follow a uniform distribution within the range of variation. Then, we generate 400 scenarios of PDN baseline loads through Monte Carlo simulation. The scenario reduction technique, such as K-means method [53], is used to downsize these generated scenarios. By calculating the distances between each generated scenario and each cluster centroid repeatedly, scenario classifications and the update of cluster centroids are achieved. The clustering process is deemed complete when no changes in the cluster compositions are observed between two consecutive iterations. We set the number of required clusters as 10 to characterize the uncertainty of PDN baseline loads in this article. In addition, the weight of each cluster centroid is determined by the number of initial generated scenarios in each cluster.
- 4) *Case IV:* In the joint optimization of bidding and pricing strategy, the EVA considers uncertainties in PDN and TN, simultaneously. The uncertainty of traffic demand is dealt with RO. We assume that the forecast error of traffic demand for each time period is $\pm 15\%$ of the predicted value. Since the EVA needs to determine the optimal bidding and pricing strategy for 11 time periods from 8:00 to 18:00, the maximum value of uncertainty budget Γ is 11 in test system 1 for

there is only a single O—D pair. Similarly, in test system 2, the maximum value of parameter Γ is 22 due to two O—D pairs. According to the conservative level of EVA towards the worst traffic demand scenario, the uncertainty budget Γ used to adjust the conservativeness of RO model is set to be 6 in test system 1 and 10 in test system 2. The maximal number of iterations is $K = 50$.

We conduct simulations from the following aspects. Firstly, in order to demonstrate the advantages of SDTA model to represent the spatial-temporal shifting characteristics of EV charging loads under time-varying charging prices compared with traditional demand elasticity model, we compare and analyze the cost/profit of EVA and the operation of PDN according to the results obtained from Case I and Case II. Secondly, by implementing Case II, Case III and Case IV, the impact of uncertainties of baseline loads in PDN and traffic demand in TN on the profit of EVA is analyzed. Thirdly, sensitivity analysis of robust SDTA model is carried out. Fourthly, a medium-scale TN-PDN test system is utilized to verify the practicability of the proposed approach. Finally, the computational performance of the proposed solution method in test system 1 and 2 is discussed.

All the simulations are carried out on the GAMS platform on a desktop computer with an Intel® Core™ i7-12700H of a 4.7 GHz CPU and 16 GB of memory. The commercial solver CPLEX is employed to solve the bi-level bidding and pricing subproblem and robust SDTA subproblem.

5.2. Comparison of bidding and pricing strategies based on demand elasticity model and SDTA model

Case I and Case II are designed to analyze the influence of SDTA model compared with demand elasticity model on the profit of EVA while bidding and pricing. It is obvious that the SDTA model is more precise than demand elasticity model to predict the charging load of each EVCS. In the demand elasticity model, the charging price is the only factor that affects the charging load. However, not only charging price, but also traffic restrictions, such as the travel time of each road, the driving route of GVs, will influence the charging decisions of EVs in the SDTA model. Due to the accuracy of SDTA model, the charging load simulated by the SDTA model is closer to reality. For the sake of simplicity, we assume that the charging load of each EVCS simulated by the SDTA model is the actual real-time charging load.

In Case I, based on the demand elasticity model, we firstly calculate the predicted charging load and the expected profit of EVA. Then the expected profit needs to be corrected. Since, there is a deviation between the day-ahead bidding quantity and real-time charging schedule, the EVA needs to sell excess electricity and purchase insufficient electricity, which will affect the final profit of EVA. In addition, according to [8], a deviation of $>5\%$ from the day-ahead bidding quantity will bring additional penalty cost to EVA. It should be mentioned that only the deviation caused by the demand elasticity model is taken into account. If there are other factors that cause changes in the actual charging load, the EVA can balance the power deviation through real-time electricity market, which is not within the scope of this article. The cost and profit allocations in two cases are shown in Table 2.

As can be seen from Table 2, the electricity purchase cost in both cases is almost the same. In addition, the expected profit of EVA in Case I is 2.11×10^3 \$ and only 60 \$ less than that in Case II. However, the

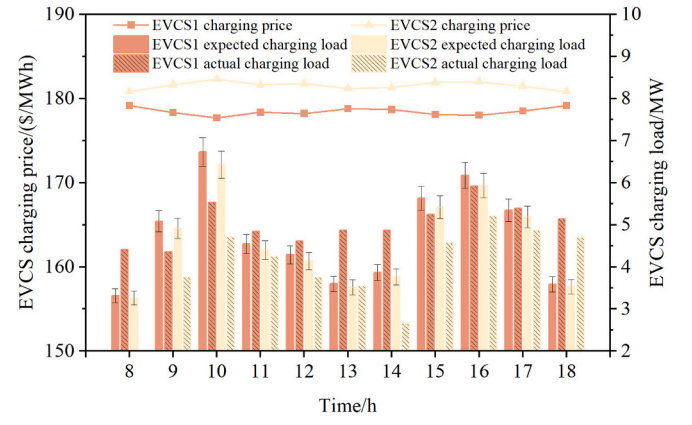


Fig. 9. Charging load and charging price of each EVCS in Case I.

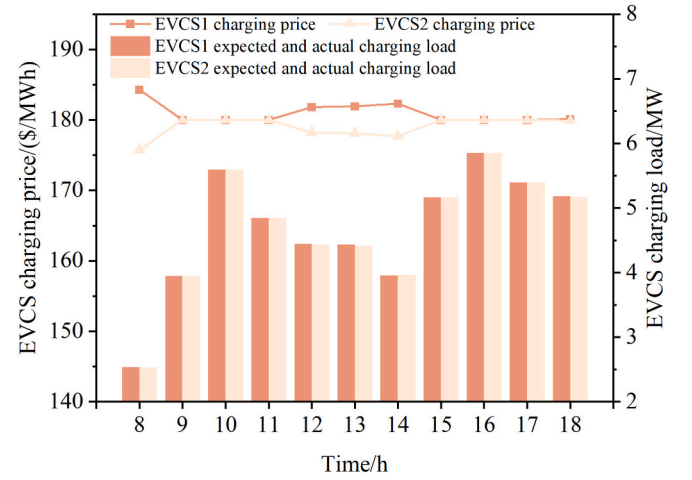


Fig. 10. Charging load and charging price of each EVCS in Case II.

expected profit of EVA in Case I needs to be corrected for there is a deviation between the day-ahead energy bids and the real-time actual charging loads. The expected charging load and charging price of each EVCS in Case I are shown in Fig. 9. Since the DLMP of EVCS 1 is lower than that of EVCS 2, the EVA will adjust the charging price to allow EVCS 1 to have more charging demand to obtain the maximum expected profit. As a result, the charging price of EVCS 1 is lower than that of EVCS 2 according to the demand elasticity model. In order to calculate the actual charging load of each EVCS, the SDTA model is utilized with the charging prices shown in Fig. 9 as parameters. The actual charging load of each EVCS is also shown in Fig. 9. It is observed that the actual charging load distribution is significantly different from the expected one. The deviation in almost every period exceeds 5% of the day-ahead bidding quantity. Thus, the EVA has to pay additional penalty cost. As a result, the final profit of EVA in Case I is 2.02×10^3 \$, lower than the expected profit of 2.11×10^3 \$.

In order to effectively increase the final profit, the EVA should utilize the SDTA model to simulate the charging decisions of EV users more precisely, then determine the bidding and pricing strategy. As shown in

Table 2
Cost and profit allocations in Case I and Case II.

	EVA			DSO		EV users
	Electricity purchase cost/ 10^3 \$	Expected Profit/ 10^3 \$	Final Profit/ 10^3 \$	Operation cost/ 10^3 \$	Network loss/MWh	Travel cost/ 10^3 \$
Case I	16.31	2.11	2.02	35.27	5.41	32.25
Case II	16.30	2.17	2.17	35.26	5.36	32.16

Table 3
Comparison on cost and profit in different cases.

Uncertainty		EVA		DSO	EV users
		Electricity purchase cost/ $10^3\$$	Profit/ $10^3\$$	Operation cost/ $10^3\$$	Travel cost/ $10^3\$$
Case II	No uncertainty	16.30	2.17	35.26	32.16
Case III	Baseline loads uncertainty	16.33	2.14	36.41	32.16
Case IV	Both uncertainties	18.24	2.27	36.72	36.57

Table 2, the final profit is consistent with the expected profit in Case II, greater than the final profit obtained in Case I. The traffic demand and PDN loads during time periods of 8:00–18:00 derived from Case II are also depicted in Fig. 5. Due to the existence of residual traffic flow, the SDTA model can reflect the temporal shifting characteristic of charging load. Hence, part of the charging load is shifted to the next time during the peak period of traffic demand (i.e., 9:00–10:00 and 15:00–16:00).

By solving the proposed model in this article, Fig. 10 shows the charging price and charging load of each EVCS. We find that the EVA adjusts the appropriate charging price so that the charging load in each time period is evenly distributed between two EVCSs to maximize profit. Specifically, during the time period when the charging load is low, i.e., 12:00–14:00, the two EVCSs will be set differentiated charging prices. When the charging load is relatively large, the charging price of two EVCSs will be the same, that is, 180 \$/MWh.

Furthermore, we evaluate the operation of PDN and TN according to the actual charging load distribution in Case I and II. As shown in Table 2, compared with Case I, the self-interested bidding and pricing behavior of EVA in Case II can improve the operation of PDN. The overall operation cost and network loss of PDN are reduced. In addition, the travel cost of EV users decreases from $32.25 \times 10^3 \$$ to $32.16 \times 10^3 \$$. Hence, the travel experience for EV users has also been improved. Since the true penetration rate of EVs is only 5%, the EVA optimal joint bidding and pricing will have a limited improvement in the operation of PDN and travel experience for EV users.

5.3. Impact of uncertainty of baseline loads in PDN and traffic demand in TN

To analyze the impact of baseline load and traffic demand uncertainties on the profit of EVA while bidding and pricing, Case II, Case III and Case IV are implemented. The cost and profit allocations considering different uncertainty scenarios are compared in Table 3.

Compared with Case II, Case III takes into account the uncertainty of baseline loads in PDN, the operation cost of PDN increases, which leads to the increase of the power purchase cost of EVA. As a result, the profit of EVA decreases slightly from $2.17 \times 10^3 \$$ to $2.14 \times 10^3 \$$. However, the uncertainty of baseline loads will not affect the formulation of the charging price of EVCSs. The EVA still makes the charging load of each period equally distributed between the two EVCSs by setting time-varying charging prices. Hence, the total travel cost of EV users remains unchanged.

The uncertainty of traffic demand has a great impact on the cost or profit of each stakeholder. When the uncertainty of traffic demand is handled by RO, the EVA needs to consider the worst scenario of traffic demand. It means that when the traffic demand reaches the upper limit of the fluctuation range, the EVA can still provide sufficient charging services, especially when the traffic demand is at a high level. With the increase of charging load, the DLMP of the PDN node connected with EVCSs shows an upward trend. Therefore, the EVA needs to pay higher electricity purchase cost. Compare with Case III, the electricity purchase cost of EVA in Case IV increases from $16.33 \times 10^3 \$$ to $18.24 \times 10^3 \$$. Similarly, with the increase of traffic demand, the operation cost of PDN

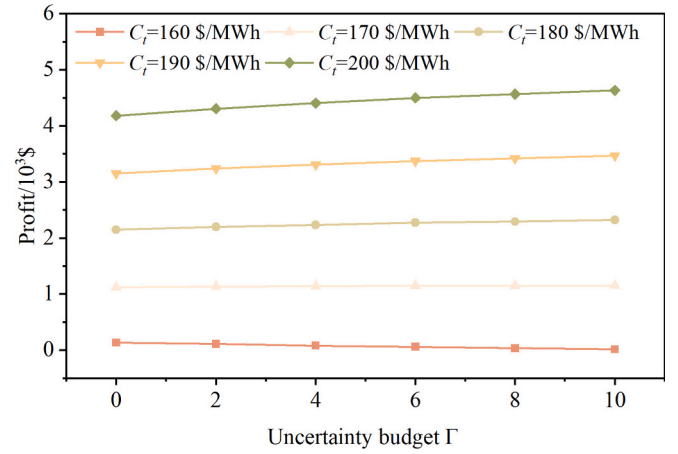


Fig. 11. Bidding and pricing profit under the influence of different average charging price level of EVCSs and uncertainty budget of robust SDTA model.

and the travel cost of EV users also increase.

The higher electricity purchase cost means that the marginal profit of electricity has decreased. However, the EVA has considered the worst scenario of traffic demand uncertainty, that is, the total charging loads have increased. Therefore, when the EVA considers the uncertainty of traffic demand in TN, they can still obtain higher profits. Whether EVA can obtain higher profits depends on the average charging price level of the two EVCSs. The impact of the average charging price level of two EVCSs will be analyzed later.

5.4. Impact of the average charging price level of EVCSs and uncertainty budget of robust SDTA model on the profit of EVA

As we mentioned before, when the EVA takes into account the uncertainty of traffic demand, whether higher profits can be obtained depends on the average charging price level. In order to analyze the impact of uncertainty of traffic demand on the EVA's profit, we calculate the bidding and pricing profit of EVA under different average charging price levels and uncertainty budgets. On one hand, the uncertainty budget Γ which reflects the conservativeness of robust SDTA model increases from 0 to 10 in steps of 2. On the other hand, the average charging price level increases from 160 \$/MWh to 200 \$/MWh with an interval of 10 \$/MWh.

This article aims at determining an optimal bidding and pricing strategy for EVA to satisfy all the charging demand in the worst traffic demand scenario. Hence, we utilize RO to deal with the uncertainty of traffic demand in TN to identify the worst scenario. The uncertainty budget Γ is used to control the level of conservatism. A larger Γ means an increase in conservatism and a worse traffic demand scenario. In addition, the traffic demand will reach the upper boundary of fluctuation in more time slots with the increase of Γ . As a result, the total traffic demand will show a growth trend with the increase in conservatism. When the traffic demand increases, whether the EVA will obtain higher profit depends on the average charging price level. Fig. 11 illustrates the trend of bidding and pricing profit of EVA with uncertainty budget Γ at different average charging price levels. In the case of our setup, since the generators at bus 1 and bus 5 cannot fully supply the baseline load in

Table 4
Comparison on cost and profit in Case IV.

EVA		DSO	EV users
Electricity purchase cost/ $10^3\$$	Profit/ $10^3\$$	Operation cost/ $10^3\$$	Travel cost/ $10^3\$$
15.47	2.48	29.01	36.19

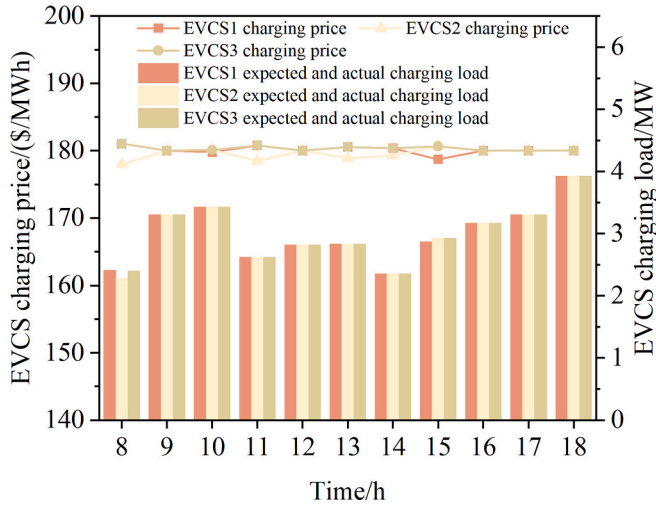


Fig. 12. Charging load and charging price of each EVCS in Case IV.

PDN as well as the charging load on the corresponding nodes, the insufficient power will be supplied by the upper grid. As a result, the DLMP of bus 1 and bus 5 is >150 \$/MWh which is the electricity purchase price of PDN from upper grid. We calculate that the DLMP of bus 1 is about 155 \$/MWh and that of bus 5 is about 165 \$/MWh. We can find that when the average charging price level of the two EVCSs is 160 \$/MWh, the profit margin of charging service provided by the EVCSs is very small, and even the EVCS 2 is in a loss state. Therefore, the profit of EVA decreases with the increase of uncertainty budget Γ .

When the average charging price level of EVCSs is >170 \$/MWh, the bidding and pricing profit of EVA shows an increasing trend with the increase of uncertainty budget Γ , and the higher the average charging level is, the more obvious the increasing trend of profit is. When the average charging price level is 170 \$/MWh, the profit is almost unchanged, only increasing from 1.12×10^3 \$ to 1.15×10^3 \$ with the increase of Γ . While the average charging price level is 200\$/MWh, the profit increases from 4.18×10^3 \$ to 4.63×10^3 \$, an increase of nearly 12%.

Thus, whether the EVA should take a conservative or radical attitude towards the uncertainty of traffic demand depends largely on the average level of charging price. When the average charging price is at a high level, the EVA should fully take the uncertainty of traffic demand into consideration, which will bring more electricity purchase cost along with more profit at the same time.

5.5. Practicability analysis on a medium-scale system

The modified 18-bus radial distribution system coupled with a 19-link transportation system is utilized to verify the practicability of the

proposed approach. There are three EVCSs considered in the test system 2. The cost and profit allocations by applying Case IV is listed in Table 4. The charging price and charging load of each EVCS can be seen in Fig. 12. As concluded in test system 1, the EVA will obtain the maximum profit when the charging load distribution is balanced among three EVCSs. The electricity purchase cost and profit of EVA are 15.47×10^3 \$ and 2.48×10^3 \$, respectively.

It is worth mentioning that even though more EVCSs are considered, the proposed method in this article is still practical. More EVCSs provide EV users with more charging options. Thus, the available paths of EVs in each O—D pair will increase. However, the number of integer variables depends on the total paths of GV and EVs. More EVCSs will introduce more integer variables into the robust SDTA model, which will affect the computational efficiency of the model to a certain extent. Hence, we need to test the convergence and efficiency of the proposed solution method and analyze the impact of the number of integer variables on the computational efficiency.

5.6. Performance evaluation of Algorithm 2 for the EVA optimal bidding and pricing

Firstly, we measure the performance of the convex relaxation and approximation methods. The error metrics representing the average percentage difference between the auxiliary variables and their replaced non-convex and nonlinear terms in test system 1 and 2 are calculated. Due to the larger scale of test system 2, the average percentage difference (i.e., $V_{\lambda}^e, V_{\lambda}^c, V_{\kappa}^e, V_{\kappa}^c$) caused by McCormick envelope relaxation method is approximately 10% in test system 1 and 12% in test system 2, respectively. While the error metrics (i.e., V_a, V_a^c) corresponding to the convex hull relaxation and piece-wise linear approximation method are relatively small and $<5\%$. In the proposed iterative solution method, the

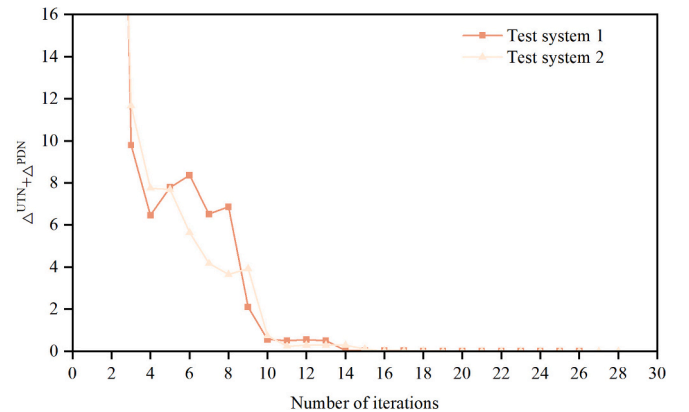


Fig. 13. Convergence performance of Algorithm 2.

Table 5

Calculation time and total number of variables of the model.

Test system	Sub-model	Number of continuous variables	Number of integer variables	Average calculation time/min	Number of iterations	Total calculation time/min
Test system 1	Bi-level bidding and pricing model	19,052	0	1.8×10^{-2}	26	214
	Robust SDTA model	u-fixed model	880	8.5		
		z-fixed model	2662			
Test system 2	Bi-level bidding and pricing model	53,713	0	8.7×10^{-2}	28	336
	Robust SDTA model	u-fixed model	2134	12.4		
		z-fixed model	7271			

decision variable sent from the robust SDTA model to the stochastic bi-level bidding and pricing model is the charging load, which is proportional to the travel flow on charging links. Due to the weak correlation between the charging load and these non-convex and nonlinear terms, the errors caused by convex relaxation and approximation methods will not have a significant impact on the profit of EVA. It is worth mentioning that several methods can be used to reduce the error metrics. For example, optimization-based bound tightening method [28] and sequential bound tightening method [30] are common ways to improve the performance of McCormick envelope and convex hull relaxation methods. In addition, the solution accuracy of piece-wise linear approximation method could be improved with the increase of the intervals, which will require much more computation time.

Then, we test the convergence and efficiency of the proposed solution method in two test systems. The solution time of Algorithm 2 and the total number of variables of each sub-model are shown in Table 5. In addition, Fig. 13 shows the convergence process of Algorithm 2 in iterations.

The solution algorithm proposed in this paper includes a bi-level iterative process. The outer iteration is to solve the bi-level bidding and pricing model and the robust SDTA model alternately. The inner iteration is to solve the u -fixed sub-model and z -fixed sub-model iteratively to obtain the worst scenario of the robust SDTA model.

As shown in Fig. 13, the proposed iterative method is capable of solving the fixed-point problem and obtaining the optimal joint bidding and pricing strategy. However, due to the large number of integer variables in the u -fixed sub-model and a bi-level loop in the solution process, the computational performance is not advantageous and it takes >3 h to obtain the optimal solution. It worth mentioning that it takes 47.1 mins to obtain the optimal bidding and pricing strategy for a single time period without considering uncertainties in [33]. Hence, the multi-period bidding and pricing strategy for EVA considering multiple uncertainties is a truly complex problem, which takes much more computation cost. As can be seen from Table 6, the computation cost required for solving the robust SDTA model is much greater than that of the bi-level bidding and pricing model. In test system 1, the average calculation time of robust SDTA model is 8.5 mins with iterations range from 2 to 4. We can find that the calculation has converged after 14 iterations and ended at 26 iterations for the gap $\Delta^{UTN} + \Delta^{PDN}$ less than the preset value ϵ . The total calculation time of solving the model established in this paper by Algorithm 2 is 214 mins. Test system 2 is more complex than test system 1. Therefore, when performing case study in test system 2, the model contains more variables and requires longer solution time. The average calculation time of robust SDTA model is 12.4 mins, and the total calculation time is 336 mins. As shown in Fig. 13, the calculation has converged after 16 iterations and ended at 28 iterations. Although the solution time in two test systems is relatively long, this article aims to discuss the bidding and pricing strategy for EVA in day-ahead spot market. Hence, the requirement for solution time is not stringent.

It is worth noting that the iteration number of test system 2 is only 2 more than that of test system 1, while the number of integer variables increases from 110 to 242. The doubling of integer variables only leads to a slight increase in the iteration number. Hence, the scalability of the proposed solution method for determining the optimal bidding and pricing strategy is shown. The more complex robust SDTA model is the main reason for the increase in calculation time. How to efficiently solve the proposed robust SDTA model is the key to improving the efficiency of the solution method, which needs further research.

6. Conclusion

In this article, in order to achieve the vehicle-grid integration from the view of EVA, the optimal joint bidding and pricing strategy for EVA considering the interactions with DSO and EV users is proposed. To handle the multiple uncertainties faced by EVA, a stochastic bi-level

model and a robust SDTA model are formulated to model the joint bidding and pricing problem and the spatial-temporal shifting characteristics of EV charging loads under time-varying charging prices, respectively. An iterative method based on fixed-point theory is designed to obtain the optimal joint bidding and pricing strategy.

Numerical results show that the proposed methodology could enable EVA to obtain higher and more substantial profit by simulating the decision-making process of DSO and EV users more practically. In addition, the EVA also has the potential to improve the operation of PDN and travel experience of EV users on the basis of fully exploring the spatial-temporal shifting characteristics of EVs. However, although the proposed iterative method can solve the optimal multi-period joint bidding and pricing problem, it has no advantage in terms of calculation time. In future works, distributed algorithm or relaxation of integer variables can be utilized to improve the computational performance. Besides, the TA model can be extended to combine the V2G technology for the EVA may achieve better economic performance by taking advantage of the flexible charging and discharging characteristics of EVs.

CRediT authorship contribution statement

Qi Wang: Writing – original draft, Software, Methodology, Conceptualization. **Chunyi Huang:** Writing – review & editing, Supervision. **Chengmin Wang:** Writing – review & editing, Validation. **Kangping Li:** Writing – review & editing, Visualization. **Ning Xie:** Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by the Key Laboratory of Modern Power System Simulation and Control & Renewable Energy Technology (Northeast Electric Power University) Open Fund (MPSS2023-10).

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